

GH - 300

GitHub Copilot

Introduction

What it is, how it works, and how to make it work well for your codebase.

What is Copilot

Architecture

Configure

Models 2026

IDE coverage

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Deck 2 · GH-300 Certified GitHub Copilot Expert Training

What is GitHub Copilot?

An AI coding assistant that helps you write code faster and with less effort, so you can focus on problem solving and collaboration.

AI code suggestions

Autocompletes lines, blocks, and entire functions in real-time using context-aware AI.

- Whole-function generation
- Comment-aware
- Test generation support

Multi-language

Works across all major languages — from web to systems programming.

- Python · JS · TS · Go
- Ruby · C++ · Java · Rust
- HTML · CSS · SQL · Shell

Deep IDE integration

Fits natively into the tools developers already use — zero workflow disruption.

- VS Code · Visual Studio
- JetBrains family
- Neovim · Eclipse · Xcode · github.com

Feature overview

Coding assistance

- Inline suggestions
- Next Edit Suggestions
- Multi-file edits in agent
- Vision (image-to-UI)
- Model picker

Quality & review

- AI code review (CCR)
- PR summaries
- Custom instructions
- AGENTS.md
- Prompt files

Agentic workflows

- Agent · Plan · Ask · Custom
- Inline Agent (Apr 2026 preview)
- Copilot CLI + /fleet
- Cloud Coding Agent
- Copilot Spaces

Enterprise & governance

- Policy management
- EMU access control
- New Usage API (Apr 2026)
- Audit logs
- BYOK / BYOR

Configuring Copilot — github.com and VS Code

On github.com

- 1 Sign in → Settings → Copilot. Click Enable or start a free trial.
- 2 Set suggestion preferences; toggle the public-code filter.
- 3 Allow / block specific repositories for Copilot.
- 4 Org owners: Settings → Copilot → Members. Enable for all or selected.
- 5 Set content-exclusion rules for sensitive repos.

In VS Code

- 1 Extensions (Ctrl+Shift+X) → install "GitHub Copilot Chat".
- 2 Click the Copilot icon in the status bar → Sign in with GitHub.
- 3 Settings (Ctrl+,) → search "GitHub Copilot" → tune per-language.
- 4 Open Chat sidebar → pick mode (Ask · Agent · Plan · Custom).
- 5 Tab to accept, Esc to dismiss, Alt+] to cycle alternatives.

Hands-on — first agent-mode prompt

Open Copilot Chat in VS Code. Switch to Agent mode. Paste this prompt. Watch what happens.

agent-mode prompt

You are a ~~vibe~~ coder using VS Code and GitHub Copilot.

I want a ~~simple~~ Todo app with this spec:

Frontend: React

Backend: Python + FastAPI

Database: SQL Server Express edition

Todo fields: ~~Id~~, Title, Description, DateCreated, IsCompleted.

Keep the code ~~simple~~ — write it like a junior developer.

Do NOT include:

- Authentication / Authorization
- Exception handling or other advanced features
- An ORM — use raw SQL instead.

Copilot architecture — three components

Copilot is a cloud-based AI coding assistant. It runs in your IDE, but the actual AI processing happens on GitHub's servers.

IDE extension

Runs locally in VS Code, Visual Studio, JetBrains. Captures context, sends requests, renders suggestions.

GitHub Cloud

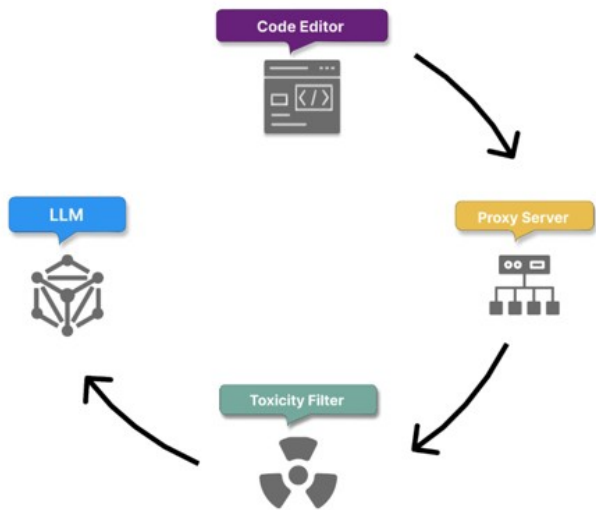
Auth + license check, content filtering, request routing, caching, telemetry. The middleware layer.

AI models (LLMs)

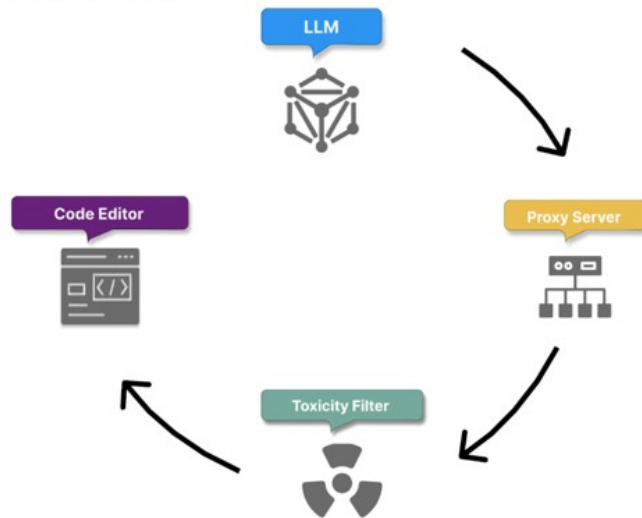
Process context + prompt.
Generate completions. Multi-model: Claude Sonnet 4.6, Opus 4.7, GPT-5, GPT-5.2-Codex, Gemini 2.5 Pro / 3.1 Pro.

Copilot architecture — Inbound and Outbound Flow

Inbound Flow



Outbound Flow



Code quality = suggestion quality

Poor naming	Good naming	What Copilot understands
<code>getD ata (x)</code>	<code>getU serById (userId)</code>	Fetches a user by ID
<code>calc(a)</code>	<code>calculateTaxW ithDiscount(amt)</code>	Calculates tax + applies discount
<code>check(str)</code>	<code>validateEm ailForm at(email)</code>	Validates email format
<code>process(x)</code>	<code>validate_em ail_address(email: str)</code>	Generates regex validation logic

Other quality signals Copilot picks up on

- Consistent patterns — if your first 5 functions follow a pattern, Copilot continues it for function 6.
- Type annotations — TypeScript and typed languages give Copilot more semantic information.
- Existing tests — Arrange-Act-Assert patterns in your suite drive matching test generation.
- Targeted comments — "Calculate total with 15% discount for orders over \$100" specifies intent.

Model lineup (August 2026) — pick by task, not by habit

Model	Strength	Use when...	Cost / 1M tokens
Claude Opus 5	Frontier reasoning; deep agentic coding	Hardest refactors; long autonomous runs.	In \$5 · Out \$25
Claude Sonnet 5	Balanced flagship; best value per token	Day-to-day coding, multi-file IDE agents.	In \$2 · Out \$10
GPT-5.6 Sol	Premium reasoning; top agentic scores	Research, large repos, hard analysis.	In \$5 · Out \$30
GPT-5.6 Terra	Near-frontier at a fraction of Sol's cost	Production workloads where cost matters.	In \$2 · Out \$12
Gemini 3.1 Pro	Long-context reasoning; multimodal	Whole-repo or doc analysis up to 1M tokens.	In \$2 · Out \$12
Gemini 3.6 Flash	Fast and cheap with 1M context	High-volume tasks, routing, classification.	In \$1.50 · Out \$7.50

Cost note: list API rates per 1M tokens (USD), August 2026. Caching cuts effective cost sharply — up to ~90% off cached reads on Claude and Gemini.

Thank you.

Open Copilot Chat. Pick a mode. Build something small.

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